Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Configuring Microsoft Entra ID SSO

You can configure Stellar Cyber to use Microsoft Entra ID SSO (formerly Azure Active Directory SSO) for **Authentication** or **Authentication and Authorization** using the procedure in this topic. The configuration includes application creation, user and group assignments, SAML setup, optional authorization, optional multi-factor authentication (MFA), and Stellar Cyber settings.

You must collect the following information as you progress through the procedure:

- User Principal Names (email addresses for users that will log into Stellar Cyber with SSO)
- Claim/Attribute Information (Applicable for Authorization only)
- Identifier (Entity ID)
- Login URL
- Reply URL (the Stellar Cyber callback URL to which Microsoft Entra ID sends its SAML authentication response)
- Certificate

The Microsoft Entra ID configuration steps are based on version 2.4.

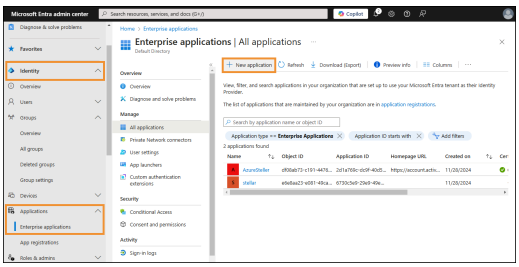
Perform the following steps in the order shown below:

1. Create a Stellar Cyber Application for SSO in Microsoft Entra ID
2. Add Users/Groups to the Stellar Cyber SSO Application
3. Configure SAML-based Single Sign-On (SSO)
4. Configure Authorization (Optional)
5. Configure Multi-Factor Authentication Support (Optional)
6. Configure Authentication in Stellar Cyber

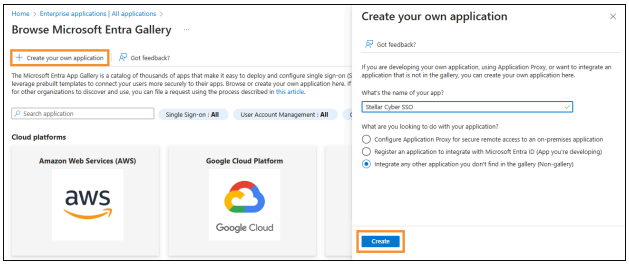
Create a Stellar Cyber Application for SSO in Microsoft Entra ID

Sign in to the Microsoft Entra Admin Center, access Enterprise Applications, and add a new application that uses Microsoft Entra ID for authentication.

1. Sign in to the Microsoft Entra Admin Center with an account that has the Cloud Application Administrator or Application Administrator role.
2. In the left-hand menu, select **Identity | Applications | Enterprise applications**.
3. At the top of the page, select **New application**.



4. In the **Browse Microsoft Entra Gallery**, select **Create your own application**.
5. Enter a name for the application (for example, **Stellar Cyber SSO**), select **Integrate any other application you don't find in the gallery (Non-gallery)**, and then **Create**.



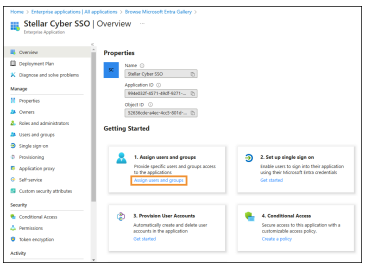
Microsoft Entra ID creates the application and displays the Overview page. With your application created, you can now add users and configure SAML.

Add Users/Groups to the Stellar Cyber SSO Application

Assign the users and groups that you want Microsoft Entra ID to authenticate to the application. You will need to add these for the following purposes:

- **For Authentication Only SSO configurations**: You will use these to create user accounts in Stellar Cyber.
- **For Authentication and Authorization SSO configurations**: You will use these to assign values to claim attributes (not applicable for per-tenant SSO).

1. In the Getting Started section of the **Stellar Cyber SSO** application Overview, select **Assign users and groups**.



   You can also access this from **Manage | Users and groups** in the side menu.
2. Select **Add user/group** in the action menu at the top of the page.
3. In the **Add Assignment** panel, select **None Selected** under **Users**.

4. Select the users or groups that you want to assign, tap or select **Select** and then **Assign** to finalize the assignments.

   For each assigned user, select its name and note the User principal name (email address). Ensure that it exactly matches the corresponding Stellar Cyber user email address. Usernames and email addresses in both Microsoft Entra ID and Stellar Cyber are case sensitive.

5. You must make note of each user's **User Principal Name** for these reasons:
   - **For Authentication Only SSO configurations**: You will use these to create user accounts in Stellar Cyber.
   - **For Authentication and Authorization SSO configurations**: For this type of configuration, users are automatically added to Stellar Cyber. You will use the User Principal Names, though to assign values to claim attributes (not applicable for Tenant-specific SSO; Authorization configurations apply only to Global SSO).
   - For *users*, select **Display Name** and copy the **User Principal Name**.
   - For *groups*, select **Display Name** and then select the **Members** link in the left side of the navigation panel. This displays a list of members (users) in that group. You can now select each user **Name** and copy the **User Principal Name**.

Now that you have added users to the application, you can configure the SSO SAML section.

Configure SAML-based Single Sign-On (SSO)

Configure basic SAML settings, download the signing certificate, and copy URLs to use when configuring SSO settings in the Stellar Cyber UI.

1. In the Getting Started section of the Stellar Cyber SSO application Overview, select **Get started** for **2. Set up single sign-on**.

   You can also access this from Manage | Single sign-on in the side menu.

2. Select the **SAML** option, select **Edit** in the **1. Basic SAML Configuration** section, fill in the required fields, and then **Save** the configuration.

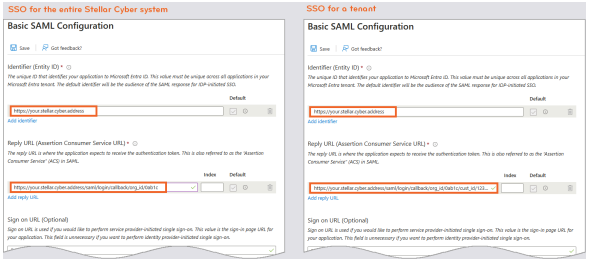
   If you are configuring SSO for the entire Stellar Cyber system:
   - **Identifier (Entity ID)**: Enter the Stellar Cyber URL. For example: `https://192.168.24.110`.
   - **Reply URL**: Use the Stellar Cyber URL, appended with `/saml/login/callback/org_id/<org-id>`. The format is `https://<your.stellar.cyber.address>/saml/login/callback/org_id/<org-id>`, where `<org-id>` is the **Organization ID** on the license page. For example: `https://192.168.24.110/saml/login/callback/org_id/0c5b67d2e9e54398be1e4d1cff1edc3`

- **Logout URL**: This setting is here for future functionality but is not currently supported.

The **Reply URL** in the Microsoft Entra ID UI is the URL to which Microsoft Entra ID sends its SAML authentication response. Different identity providers use different names for this URL as shown in the following table:

If you are configuring SSO for a tenant:

- **Identifier (Entity ID)**: Enter the Stellar Cyber URL.
- **Reply URL**: Use the Stellar Cyber Platform URL, appended with `/saml/login/callback/org_id/<org id>/cust_id/<tenant-id>`. The format is `https://<your.stellar.cyber.address>/saml/login/callback/org_id/<org-id>/cust_id/<tenant-id>`, where `<org-id>` is the **Organization ID** on the license page. For example: `https://192.168.24.110/saml/login/callback/org_id/0c5b67d2e9e54398be1e4d1cff1edc3d/cust_id/59125044`
- **Logout URL**: This setting is here for future functionality but is not currently supported.



The **Reply URL** in the Microsoft Entra ID UI is the URL to which Microsoft Entra ID sends its SAML authentication response. Different identity providers use different names for this URL as shown in the following table:

Identity Provider	Term for the Assertion Consumer Service (ACS) URL
Active Directory Federation Services	Relying party SAML 2.0 SSO service URL
Microsoft Entra ID	Reply URL
Okta	Single sign on URL
OneLogin	Assertion Consumer (ACS) URL
Rippling	Assertion Consumer Service URL

Configure Authorization (Optional)

Authorization allows you to assign Stellar Cyber scopes, privileges, and tenant options within Microsoft Entra ID, rather than within Stellar Cyber. If you want to configure Authorization in addition to Authentication, you must add Microsoft Entra ID attributes / claims (scopes) for use in Stellar Cyber This procedure adds the required fields to the Microsoft Entra ID users you associated with the application.

You must configure the basic identity, above, before you perform the steps in this section. If you are configuring **Authentication** only, or per-tenant SSO authentication, skip this section.

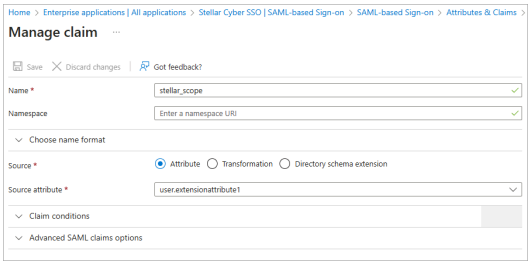
Add Stellar Cyber Fields to User Accounts

1. In the **Set up Single Sign-On with SAML** section of the Stellar Cyber SSO application configuration, select **Edit** in the **2 Attributes & Claims** section.
2. Select **Add new claim**, enter the following claim settings, leave the others as they are, and then **Save** the configuration.

   Name: Enter `stellar_scope`.

   Source: **Attribute**

   Source Attribute: Choose a relevant user attribute (for example, `user.extenstionattribute1`).



You define attributes in this step; values are added in a later step.

3. Repeat the previous step to add the following attributes: `stellar_privilege`, `stellar_tenant`, and `stellar_tenant_group`

Take care when entering the required attributes and values. Typos in either mean that users are not authenticated and cannot log in. Typos in the optional attributes and values mean that users are not assigned to the appropriate tenant or tenant group (but are authenticated).

CustomAttributeName	Source Attribute*	Values (setlater)	Global SSO	Tenant-specific SSO
stellar_scope	user.extensionattribute1	root partner tenant	Required for **Authorization**	Not applicable
stellar_pri	user.extensionattribute2	super_admin	Required for **Authorization**	

vilege		platform_admin security_admin user	
stellar_tena	user.extensionattribute3	ID number for conf	(Optional) Specify an individual tenant ID, not name. The ID is available on the Tenants List page.

stellar_tenant_group	user.extensionattribute4	configured tenant group ID number for any configured tenant group	(Optional) Specify a tenant group ID, not name. This is typically available for use by MSSP users with the **Partner** role. The Tenant Group ID is displayed on the Tenant Groups page.

Any available user.extensionattributeX (where X is 1-15) may be used.

> Values in these fields are case sensitive and syntax matters. Use the exact indicated syntax and verify that you have made no typos. If you have created a custom privilege with spaces or dashes, use an underscore instead. Example: A

custom privilege of **STML-Security Admin** must be entered as `STML_Security_Admin`.

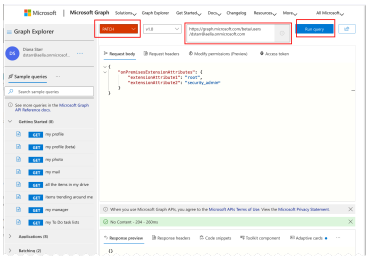
4. When done, exit the **Attributes & Claims** editor panel.

Assign Values to Stellar Cyber Fields (Graph Explorer Method)

Microsoft Entra ID does not provide a GUI-based mechanism to modify the attribute values. This procedure uses Microsoft Graph Explorer to edit and validate the attributes. If you wish to automate the steps, refer to the PowerShell method, below.

1. To set the attribute values, retrieve the list of **User Principal Names** you saved above.
2. Open Microsoft Graph and log in with your **administrator** Microsoft Entra IDCredentials (https://developer.microsoft.com/en-us/graph/graph-explorer).
3. In the query pane, change the **GET** action menu to **PATCH**.
4. Now enter this query URL, replacing the **<user principal name>** below with a User Principal Name you saved earlier.

   `https://graph.microsoft.com/beta/users/<user principal name>`



5. Enter the following content as the **Request Body**, replacing the attribute values with one of the supported values in the table above:

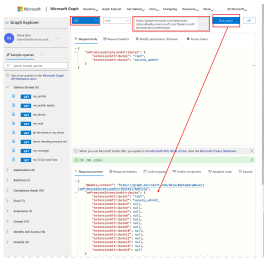
   Take care when entering the required attributes and values. Typos in either mean that users are not authenticated and cannot log in. Typos in the optional attributes and values mean that users are not assigned to the appropriate Stellar Cyber tenant or tenant group (but are authenticated).

```
{
"onPremisesExtensionAttributes": {
     "extensionAttribute1": "root",
     "extensionAttribute2": "security_admin"
     }
}
```

6. Click **Run Query** to set the values.
7. To verify the values were set correctly, change the query type back to **GET**.
8. Now enter this query URL, replacing the **<user principal name>** below with the same User Principal Name (Microsoft Entra ID login) you just set above.

   `https://graph.microsoft.com/beta/users/<user principal name>?$select=onPremisesExtensionAttributes`

9. Click **Run Query** and review the list of extensionAttributes to verify the values were set.



Assign Values to Stellar Cyber Fields (PowerShell Method)

- **For Authentication Only SSO**: First, manually add all users on the Stellar Cyber Platform. After this manual entry, users can log in with SSO. Use the list of User Principal Names that you saved.
- **For Authentication & Authorization SSO**: Configure all users through the IdP.
- **For Local access (bypass) when SSO is enabled**: If Stellar Cyber loses connectivity with your IdP, users configured for SSO cannot log in. As a preventive bypass method, manually create a new user in the Stellar Cyber Platform with root scope and with a valid email address that has "`+admin`" appended to a valid user name, as follows: `<user>+admin@yourorganization.com` (`joe+admin@yourorganization.com`). The user you create must be able to receive a password reset email at `<user>@yourorganization.com`. This email alias is what Stellar Cyber uses to permit the bypass of an SSO for a local login. After you create this separate user account, the user can log in two ways:
    - An SSO user with `<user>@yourorganization.com`

      or
    - A local user at `https://your.stellar.cyber.address/login` using `<user>+admin@yourorganization.com`

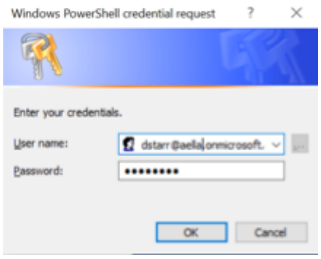
If SSO is configured, it's recommended to keep an active administrative account in the Stellar Cyber user management.

As an alternative to the Microsoft Graph Explorer method, you can use PowerShell to set the attribute values. The commands in this section can also be automated in PowerShell.

1. Open **PowerShell** from your Windows system **as an administrative user.**
2. From the PowerShell command line, run the bolded commands and replace **$User** value with the login you used above. The last two lines illustrate assigning values to the stellar_scope and stellar_privilege attributes. If you used all four attributes, add commands for those and specify the Tenant ID and Tenant Group ID exactly as they are listed in Stellar Cyber.

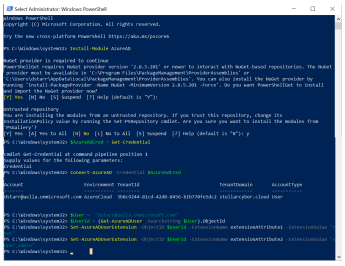
Be sure that you assign the correct value to the correct extension number. In this example extensionAttribute1 is for stellar_scope, which is being given root; extensionAttribute2 is for stellar_privilege and is being assigned security_admin. Review your list of extension attributes carefully before running these commands.

```
PS C:\WINDOWS\system32> Install-Module AzureAD
Untrusted repository
You are installing the modules from an untrusted repository. If you trust
this repository, change its
InstallationPolicy value by running the Set-PSRepository cmdlet. Are you sure
you want to install the modules from
'PSGallery'?
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help
(default is "N"): Y
PS C:\WINDOWS\system32> $AzureAdCred = Get-Credential
Enter your Azure admin user and password
```




```
PS C:\WINDOWS\system32> Connect-AzureAD -Credential $AzureAdCred
PS C:\WINDOWS\system32> $User = "dstarr@aella.onmicrosoft.com"
PS C:\WINDOWS\system32> $UserId = (Get-AzureADUser -Searchstring
$User).ObjectId
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> Set-AzureADUserExtension -ObjectID $UserId -
ExtensionName extensionAttribute1 -ExtensionValue "root"
PS C:\WINDOWS\system32> Set-AzureADUserExtension -ObjectId $UserId -
ExtensionName extensionAttribute2 -ExtensionValue "security_admin"
PS C:\WINDOWS\system32> exit
```

Following is an example of this sequence:



Configure Multi-Factor Authentication Support (Optional)

Microsoft Entra ID and Stellar Cyber support multi-factor authentication (MFA). If you intend to set up Stellar Cyber access with two-factor authentication (2FA; a subset of MFA), you must enable a **Conditional Policy** in Microsoft Entra ID that supports this type of authentication and assign it to users and groups.

Configuration of MFA in Microsoft Entra ID is independent of whether you enable 2FA in Stellar Cyber. It is possible for you require users to authenticate with MFA in Microsoft Entra ID and again with 2FA in Stellar Cyber

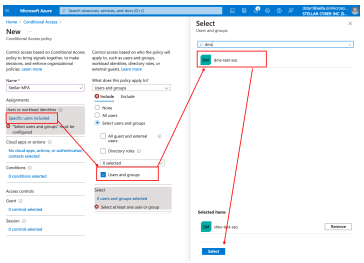
1. In Microsoft Entra Admin Center, select **Protection | Security Center | Conditional Access**.

   If you don't see Security Center in the Protection section of the menu, select **Show more**.

2. Select **+ Create new policy** and enter a name for the policy, such as **Require MFA for Stellar Cyber**.

3. Under **Assignments**, select the **Specific users included** link for **Users or workload identities** and then select **Specific users included**.

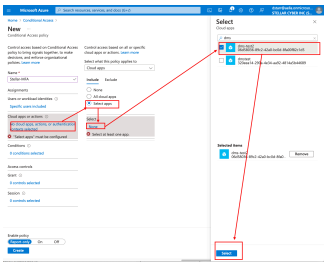
4. Select the **Users and groups** that you specified to be members of the Stellar Cyber Microsoft Entra ID application created above, and then select **Select**.



5. Under **Assignments**, select the **No cloud apps, actions, or authentication contexts selected** link for **Cloud apps or actions**.

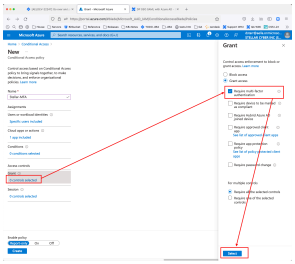
6. Select **Include | Select apps**, and then select the **None** link to open the **Select Cloud apps** panel.

7. When the selection pane opens, locate the Stellar Cyber Microsoft Entra ID application you created above and then select **Select**.



8. In the **Grant** section, select **0 controls selected**.

9. In the side panel that opens, select **Grant access**, select **Require multi-factor authentication**, and then select **Select**.



10. At the bottom of the page, change the **Enable policy** toggle from **Report-only** to **On**.

11. Select **Create**.

    The policy is created and the **Conditional Access Policy** page is redisplayed to include the new policy.

Collect Microsoft Entra ID Access Details

At this stage, you should have collected all your user information and have set up the Microsoft Entra ID application for Stellar Cyber SSO access. Your certificate and access details are now ready to copy/download. Use this procedure to ensure you have all the noted information before you proceed to the next section.

If you are configuring **Authorization**, you must perform that procedure (see previous section) prior to downloading the certificate in this section.

1. If you have navigated away from the application page, select **Home** on the portal and navigate back **Identity | Applications | Enterprise applications**.
2. Locate and display the application you created above.
3. Select **Single sign-on** from the left side navigation or select **Get started** for **2. Set up single sign on**.
4. On the **Set up Single Sign-On with SAML** page, download or copy (as appropriate) the items below for use when you set up SSO in Stellar Cyber:
    - **Step 1: Basic SAML Configuration**

      Identifier (Entity ID) – The Stellar Cyber URL. Microsoft Entra ID uses this to recognize and establish a trust relationship with Stellar Cyber as a service provider (SP).
    - **Step 2: Attributes & Claims** (Applicable only if you configured Authorization)

      `User.extensionattributeX` values for all the Stellar Cyber claim names you added
    - **Step 3: SAML Certificates**

      Certificate (base 64) – You upload this as the IdP certificate in Stellar Cyber.
    - **Step 4: Set up Stellar Cyber SSO**

      Login URL – This is used for the **Entry Point URL** field in Stellar Cyber, to link with Microsoft Entra ID.

In addition to the above details, you should have previously noted all of the **User Principal Names** (email addresses) for all users that will use Microsoft Entra ID SSO with Stellar Cyber.

Configure Authentication in Stellar Cyber

With all your details collected, you are now ready to configure SSO Authentication in Stellar Cyber. The steps below are generally applicable for use for global configuration or per-tenant configuration.

Prepare for Users

- **For Authentication Only SSO**: First, manually add all users on the Stellar Cyber Platform. After this manual entry, users can log in with SSO. Use the list of User Principal Names that you saved.
- **For Authentication & Authorization SSO**: Configure all users through the IdP.

- **For Local access (bypass) when SSO is enabled**: If Stellar Cyber loses connectivity with your IdP, users configured for SSO cannot log in. As a preventive bypass method, manually create a new user in the Stellar Cyber Platform with root scope and with a valid email address that has "`+admin`" appended to a valid user name, as follows: `<user>+admin@yourorganization.com` (`joe+admin@yourorganization.com`). The user you create must be able to receive a password reset email at `<user>@yourorganization.com`. This email alias is what Stellar Cyber uses to permit the bypass of an SSO for a local login. After you create this separate user account, the user can log in two ways:
    - An SSO user with `<user>@yourorganization.com`

      or
    - A local user at `https://your.stellar.cyber.address/login` using `<user>+admin@yourorganization.com`



If SSO is configured, it's recommended to keep an active administrative account in the Stellar Cyber user management.

Enable SSO

1. Log in to Stellar Cyber and select **System | ORGANIZATION MANAGEMENT | Settings**.
2. In the **Authentication Settings** section, select **SSO (SAML)** as the **Authentication Method**.
3. Select **Metadata URL** or **Manual Config**:

   If you select **Metadata URL**:

   - **Issuer URL**: If this setting is present, enter the Stellar Cyber URL.
   - **Metadata URL**: Paste the URL you copied from the IdP. The URL must begin with either `https://` or `http://`. Different vendors use different names for the Metadata URL:

Identity Provider	Term for Metadata URL
Active Directory Federation Services	Federation Metadata URL
Microsoft Entra ID	App Federation Metadata URL
Okta	Metadata URL
OneLogin	Issuer URL
Rippling	Metadata URL

   If you select **Manual Config**:

   - **Issuer URL**: If this setting is present, enter the Stellar Cyber URL.
   - **Entry Point**: Paste the entry point URL you copied from the IdP.

     You must include **http://** or **https://** before the URLs. Different vendors use different names:

Identity Provider	Term for Entry Point URL
Active Directory Federation Services	SAML Endpoint
Microsoft Entra ID	Login URL or SAML-P Sign-on Endpoint
Okta	Sign on URL
OneLogin	SAML 2.0 Endpoint (HTTP)
Rippling	SAML 2.0 Endpoint (HTTP)

- **IDP Certificate**: Upload the certificate file you downloaded from the IdP.

4. Select **Require Signed Response** only if the IdP requires it.



Verify whether your IdP requires signed SAML responses. Some IdPs, such as Okta, require them by default. If so, then enable **Require Signed Response** here as well.

5. Select **Allow Clock Skew** to allow for system time differences between Stellar Cyber and your IdP. Authentication messages have an expiration. If the system times on Stellar Cyber and your IdP are not synchronized, the messages might expire before they even get to Stellar Cyber. The result is that users cannot log in, because they cannot authenticate.
6. Choose your IdP setting: **Authentication Only** or **Authentication and Authorization**. Note the following:



- A Global selection of **Authentication and Authorization** applies to all users (root, partner, and tenant), so the option to change authentication method for a specific tenant is not applicable when the Global method is set to **Authentication and Authorization**. You cannot log in to Tenant SSO when Global SSO is set to **Authentication and Authorization**. If you want to use SSO but also allow local users and tenant override, you must set the Global authentication method either to **Local** or to use the IdP with **Authentication Only**.
- Although you can customize SSO configuration on a per-tenant basis, the **Authorization** capability is only supported at the global level. Overrides you make at the tenant level are for **Authentication** only, so the toggle for **Authentication and Authorization** is not offered in the tenant editor.
- The authentication method for partners is the same as that for root users. Any authentication overrides for tenant-level users in a tenant group have no effect on the authentication method for the partners who manage the group.

- Choose **Authentication Only** for Stellar Cyber to authenticate users from your IdP, but manage scope and privilege locally. If you choose this, you must create the user in Stellar Cyber before adding them to your IdP.
- Choose **Authentication and Authorization** for Stellar Cyber to authenticate users from your IdP, along with their scope and level of privilege. You must

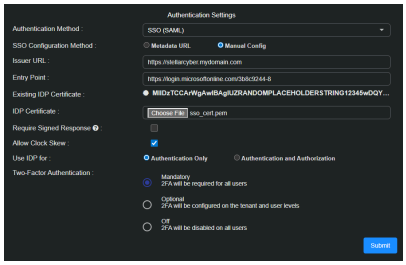
configure authorization on your IdP before enabling this, otherwise users cannot log in. If you choose this, you do not need to create the user in Stellar Cyber. Stellar Cyber creates the user and assigns scope and privilege based on the information passed from the IdP.

> When Global Settings is configured for **Authentication & Authorization**, the option to **Create** new users manually is hidden because new users must come from the IdP source.

7. Choose a **Two-Factor Authentication** option that matches your IdP configuration:
    - **Off**: If you choose this option, Stellar Cyber user accounts are not offered a 2FA option.
    - **Mandatory**: If you choose this option, all users for every tenant are required to use 2FA when logging in to Stellar Cyber.
    - **Optional**: If you choose this option:
        - The 2FA option can be customized for individual tenants under **System | ORGANIZATION MANAGEMENT | Tenants**.
        - Individual users can choose to enable 2FA under their **User Profile** in the Stellar Cyber UI.
        - You can enforce 2FA for specific users under **System | ORGANIZATION MANAGEMENT | Users** when adding or editing a user

> - The overall Global Settings for 2FA affect authentication for partners and tenant users. For example, if 2FA is **Mandatory**, all users must use 2FA.
> - Enabling 2FA here is independent of what you have configured on your SSO service. Enabling it here causes a separate 2FA prompt to be displayed upon logging in to Stellar Cyber.
> - The 2FA page from Stellar Cyber refers to use of Google Authenticator, but other authenticator applications also work.

8. Review your settings and then **Submit** them.



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